

Images mapped to calculator products (PANELS_TECHO) + cross-ref to internal TECHMET/BMC DWGs

Product Image Mapping — Verification

Collected visual references mapped to exact products in the BMC Calculadora (src/data/constants.js PANELS_TECHO).

For verification that renders match real manufactured panels (Kingspan Uruguay / Bromyros / BMC). Includes cross-reference to recently inspected internal DWGs (TECHMET constructive details + BMC accessory profiles).

Companion files: [PRODUCT-IMAGE-MAPPING-VERIFICATION.html](#) (interactive) | [MATERIAL-VISUAL-SELECCION-LIO.html](#) | [knowledge/INTERNAL-TECHMET-BMC-DWG-CONSTRUCTIVE-DETAILS-INVESTIGATION.md](#) | [knowledge/ISOROOF-PLEGADOS-REAL-REFERENCES-UY-2026.md](#)

Quick Mapping Table: Image → Calculator Product

Local Image / PDF	Calculator Product	Key Visual Match
isoroof-plus.png	ISOROOF_PLUS (ISOROOF PLUS 3G)	3 grecas upper + imitation wood interior. 30/50/80mm, au=1.0m, ext white/grey/red
isoroof-3g.png	ISOROOF_3G (ISOROOF 3G)	Standard trapezoidal 3G profile, white interior. 30-100mm, au=1.0m
isoroof-foil.png	ISOROOF_FOIL (ISOROOF FOIL 3G)	Same 3G profile + flexible white foil interior
isoroof-colonial.jpg	ISOROOF_COLONIAL	Colonial tile aesthetic exterior + white interior. au=1.0m
isodec-pir.png	ISODEC_PIR (ISODEC PIR) + EPS ref	Engrafado joint system. au=1.12m, 50/80/120mm PIR (EPS variants too)
bmcuguay-isoroof-3g-*.jpg/png (x3)	ISOROOF_3G (and Isoroof family)	Commercial photos from BMC Uruguay distributor. Exact 'Trapezoidal 3G' sold items
chacras-del-sur-cubierta-roja.jpg + other obra photos	ISOROOF_3G / PLUS (real installed roofs)	Real UY installed examples showing plegados/grecas in context (red, industrial, residential)
isoroof-plus-bromyros-ficha.pdf + bromyros-isodec-ficha.pdf	All Isoroof + Isodec	Official Bromyros tech sheets with explicit 'trapezoidal (3 grecas)' text + specs

Note: The obra and bmcuguay images show the real-world appearance of the exact products already in the calculator. The Bromyros PDFs + the recently inspected internal DWGs (TECHMET series + BMC frontales/babetas/desarrollos) provide the precise plegado geometry and accessory profiles (forros F1-MET etc.) that match the 3-grecas trapezoidal upper sheet.

Images mapped to calculator products (PANELS_TECHO) + cross-ref to internal TECHMET/BMC DWGs

ISOROOF PLUS 3G — ISOROOF_PLUS

Techos Premium • au=1.0m • 30/50/80mm • PIR • 3 grecas upper + imitation wood interior • Ext colors: white/grey/red

COEFICIENTES TÉRMICOS

ESPEJOR DEL PANEL (mm)	K (w/m2K)	R (m2K/W)
30	0,733	1,364
50	0,440	2,273
80	0,275	3,636

Why it maps perfectly: Explicit manufacturer text: 'cara superior con 3 grecas como terminación e interior con acero galvanizado' + 'cálida terminación interior de imitación madera'. Fabricado en Planta Kingspan Uruguay. Bromyros PDF confirms 'exterior en forma trapezoidal (3 grecas)'.

DWG cross-ref (internal TECHMET/BMC files): MET-TECHMET-01 (PLEGADO DOBLE), MET-07 & 09-12 (FORRO F1-MET-1 to F5 + FORRO LATERAL/ATRAQUE, 'ANCHO UTIL 1000 mm', 1@30cm fixings, 'CALZAR PANEL TECHMET'). The forros F*-MET codes are the exact accessory plegados that go with this panel's 3-greca profile.

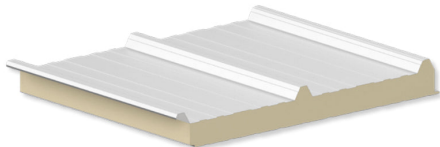
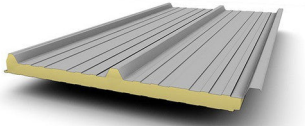
Use for verification / viz: Primary reference for 3D upper chapa profile extrusion (exact 3-greca trapezoid matching your plegados-en-general.dwg), exterior metal albedo + interior wood PBR, real colors. Use for PanelCrossSection layer stack + hatching, and PDF appendix sections.

ISOROOF 3G — ISOROOF_3G

Techos Livianos • au=1.0m • 30/40/50/80/100mm • PIR • Standard trapezoidal 3G profile • Interior white

COEFICIENTES TÉRMICOS

ESPEJOR DEL PANEL (mm)	K (w/m2K)	R (m2K/W)
30	0,733	1,364
40	0,550	1,818
50	0,440	2,273
80	0,275	3,636
100	0,220	4,545



Why it maps perfectly: Base 'Trapezoidal 3G' profile of the Isoroof family. Multiple thicknesses match calculator. 'Fabricado en Planta Kingspan Uruguay'. bmcuguay commercial shots are the exact visuals of the product sold/cotized (specs: chapa 0.55/0.45mm, au 1m, 'Trapezoidal 3G').

DWG cross-ref (internal TECHMET/BMC files): MET-07/09-12 show identical 'PANEL TECHMET' with the 3G profile + lateral/atracque forros. Desarrollos.dwg (and your plegados-en-general.dwg, same AC1032 version) give the precise flat-to-3D plegado geometry for the upper grecas.

Use for verification / viz: Standard 3G profile for 3D mesh extrusion, prepainted metal textures, joint details. bmcuguay shots for commercial/real-sold appearance validation. Good base for color variants and PDF 'Sección 2D'.

Images mapped to calculator products (PANELS_TECHO) + cross-ref to internal TECHMET/BMC DWGs

ISOROOF FOIL 3G — ISOROOF_FOIL

au=1.0m • 30/40/50/80mm • PIR • Identical 3G profile + flexible white foil interior liner

COEFICIENTES TÉRMICOS

ESPESOR DEL PANEL (mm)	K (w/m2K)	R (m2K/W)
30	0,697	1,435
40	0,535	1,87
50	0,434	2,304
80	0,277	3,609

Why it maps perfectly: Same trapezoidal 3G upper profile as the 3G/Plus family, differentiated only by interior finish (foil). Specs (au, espesores, colors) identical to ISOROOF_3G.

DWG cross-ref (internal TECHMET/BMC files): Same family of forros and plegados as the other Isorooft 3G items in the TECHMET DWGs. The 'Desarrollos' give the common greca geometry; only the interior liner changes.

Use for verification / viz: Variant for interior material (foil sheen/roughness in PBR). Upper profile geometry is identical to ISOROOF_3G — reuse the same extrusion code with different bottom face material.

Images mapped to calculator products (PANELS_TECO) + cross-ref to internal TECHMET/BMC DWGs

ISOROOF COLONIAL — ISOROOF_COLONIAL

au=1.0m • PIR • Colonial tile aesthetic on exterior + white interior (distinct finish)



Why it maps perfectly: 'Diferencial estético' with colonial tile exterior. au 1000mm, PIR. White interior. Listed on the UY site and in your calculator even if one source notes Brazil plant origin.

DWG cross-ref (internal TECHMET/BMC files): Shares the base Isorooft trapezoidal family geometry (forros and plegados from the TECHMET series still apply; the exterior 'teja' is the differentiating finish).

Use for verification / viz: Aesthetic variant for exterior. Base profile compatible with the 3G family — useful for showing finish options in the family showcase / visor.

ISODEC PIR (and EPS reference) — ISODEC_PIR

Techos y Cubiertas • au=1.12m • 50/80/120mm PIR (EPS up to 250mm) • Engrafado joint system

COEFICIENTES TÉRMICOS

ESPESOR DEL PANEL (mm)	K (w/m2K)	R (m2K/W)
50	0,44	2,27
80	0,275	3,636
120	0,183	5,464

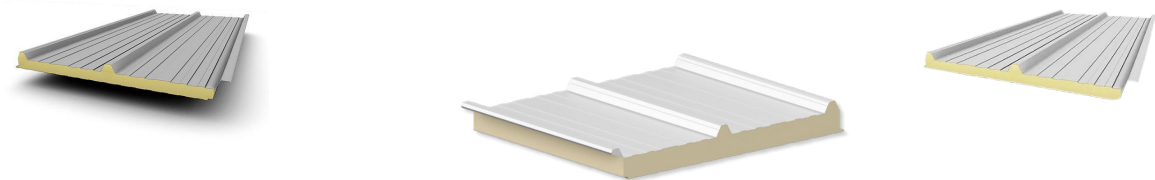
Why it maps perfectly: Explicit 'sistema de unión engrafado entre paneles'. au 1120mm exact match. Thicknesses and PIR/EPS core match calculator. 'Lugar de fabricación: Planta Kingspan Uruguay / Canelones Bromyros'.

DWG cross-ref (internal TECHMET/BMC files): MET-16 references e=50mm panels and compatible metal structure/angles. The BMC accessory DWGs (BabetaLateral, FRONTALES) are used across both Isorooft and Isodec families for trims and borders.

Use for verification / viz: Different joint detail (engrafado vs greca borders) for 3D side geometry and encounters. Standard texture/color reference. The Bromyros Isodec ficha PDF gives the layer construction for 2D sections.

Commercial / Real-Sold Images from BMC Uruguay Distributor (bmcuruguay.com.uy)

These are the actual photos of the products that are listed, priced, and sold through the BMC channel that feeds the calculator. They show the real commercial appearance (lighting, scale, finish) of ISOROOF 3G and family.



Images mapped to calculator products (PANELS_TECHO) + cross-ref to internal TECHMET/BMC DWGs

Source: <https://bmcuругuay.com.uy/products/isoroof-3g-gris-rojo-blanco-bromyros> (and /collections/isoroof). Text on site explicitly says 'Trapezoidal 3G', chapa calibre 24/26, au=1m, fabricación nacional. Accessories on the site reference 'grecas del Panel Isoroof'. Perfect commercial reference for the exact SKUs in your PANELS_TECHO.

Real Installed Examples (Obras) — plegados/grecas in context

Photos from kingspan.com.uy/fotos-de-obras showing actual installed roofs in Uruguay. These are the best references for how the 3-greca plegados look at scale, under real light, with joints, fixings, and surrounding architecture.



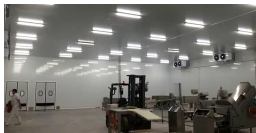
Chacras del Sur — red roof (excellent plegados visibility)



Techo vivienda particular (close detail)



Vivienda en paneles (residential scale)



Frigorífico Sarubbi (large industrial)



Planta Bromyros (manufacturing context)

Full set (50+): <https://kingspan.com.uy/fotos-de-obras/>. Use for final visual validation of 3D renders and any 'real obra' appendix pages in PDFs.

Cross-Reference: Internal DWGs (TECHMET + BMC Accessories) — Geometry Source

The raster images above show appearance and finishes. The recently inspected internal DWGs provide the precise vector geometry for the plegados, forros, and accessories that go with these panels:

- **TECHMET series (MET-TECHMET-*.dwg)**, especially MET-07 and the 09-10-11-12 combined: 'FORRO F1-MET-1' through F6, FORRO LATERAL, FORRO ATRAQUE, 'PANEL TECHMET', 'ANCHO UTIL 1000 mm', 30/50/80 mm, 1@30 cm fixings, 'PLEGADO DOBLE', 'CALZAR PANEL', 'TORNILLO AUTOPERFORANTE AL MONTE'. These are the exact constructive details for the flashings that accompany the 3-greca Isoroof panels.
- **Desarrollos.dwg** (and your earlier plegados-en-general.dwg — same AC1032 version): flat pattern developments for fabricating the plegados/grecas. Best source for exact bend geometry, leg lengths, and unfoldings.
- **BabetaLateral.dwg + FRONTALES.dwg / FRONTALNuevo.dwg**: the lateral and frontal trim profiles (babeta, gotero frontal/greca, etc.) used on the panel edges. Pure profile geometry matching your BORDER_OPTIONS and PERFIL_TECHO entries.
- All files live in your Dropbox 'bmc - interna' and 'BMC - Uruguay' paths. Open them in CAD/FreeCAD alongside the images above to trace/export the profiles for the 3D code and 2D CAD sections.

Technical PDFs in the bundle (for layer details and specs): [isoroof-plus-bromyros-ficha.pdf](#) and [bromyros-isodec-ficha.pdf](#). These contain the 'trapezoidal (3 grecas)' confirmation and full thermal/constructive tables.

PDF generated 2026-06-02 02:12 from bundled assets in material-visual-isoroof-uy/.

All images and PDFs are official public references from kingspan.com.uy, bmcuругuay.com.uy, and Bromyros. The internal DWGs are company assets for geometry reference only.

This document + the companion HTML + the two knowledge/ reports = the complete visual + CAD reference set for making 2D/3D viz match the real products already in the calculator.